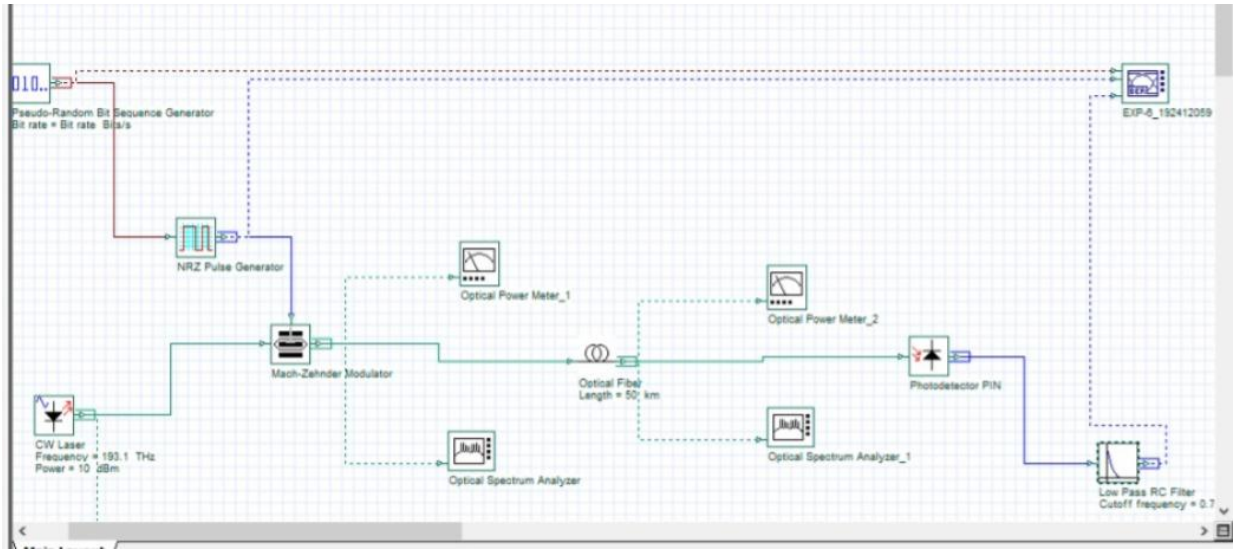
EXPERIMENT 16:
Fiber-Optic Link Design, Link Budget and BER Analysis Using OptiSystem

OBJECTIVES :

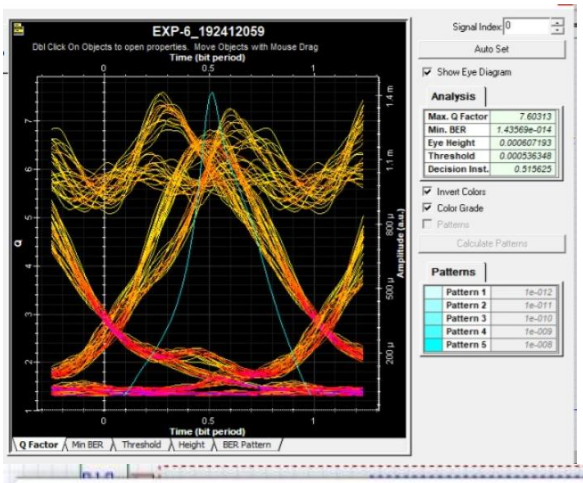
- To design and simulate a fiber-optic communication link using OptiSystem.
- To study the transmission of optical signals through optical fiber.
- To analyse the link budget and calculate the power losses.
- To evaluate the Bit Error Rate (BER) of the received signal.
- To study the effects of attenuation and noise on communication performance.

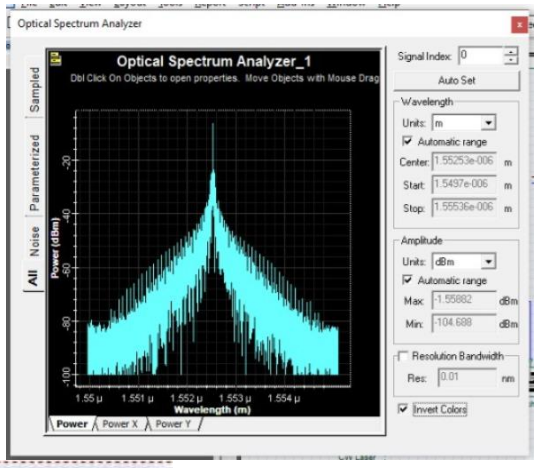
SOFTWARE REQUIRED: Opti System.

CIRCUIT DIAGRAM:

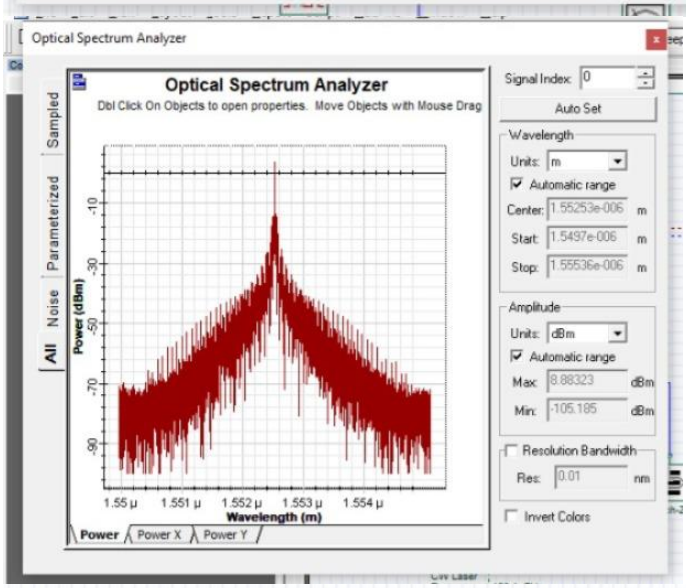


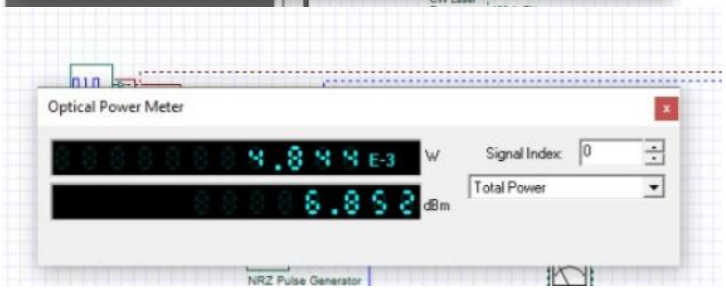
OUTPUTS:


EXP-6_192412059
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Signal Index 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 7.60313
Min. BER 1.43569e-014
Eye Height 0.000607193
Threshold 0.000536348
Decision Inst. 0.515625
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008
Q Factor
Min BER
Threshold
Height
BER Pattern


Optical Spectrum Analyzer
Optical Spectrum Analyzer_1
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Sampled
Parameterized
All
Noise
Power (dBm)
Wavelength (m)
Signal Index 0
Auto Set
Wavelength
Units: m
Automatic range
Center: 1.55253e-006 m
Start: 1.5497e-006 m
Stop: 1.55536e-006 m
Amplitude
Units: dBm
Automatic range
Max: -1.55882 dBm
Min: -104.688 dBm
Resolution Bandwidth
Res: 0.01 nm
Invert Colors
Power
Power X
Power Y


Optical Power Meter
484.406 E-6 W
-3.148 dBm
Signal Index 0
Total Power


Optical Spectrum Analyzer
Optical Spectrum Analyzer
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Sampled
Parameterized
All
Noise
Power (dBm)
Wavelength (m)
Signal Index 0
Auto Set
Wavelength
Units: m
Automatic range
Center: 1.55253e-006 m
Start: 1.5497e-006 m
Stop: 1.55536e-006 m
Amplitude
Units: dBm
Automatic range
Max: 8.88323 dBm
Min: -105.185 dBm
Resolution Bandwidth
Res: 0.01 nm
Invert Colors
Power
Power X
Power Y


Optical Power Meter
4.844 E-3 W
6.852 dBm
Signal Index 0
Total Power
NRZ Pulse Generator

REFLECTION:

- Learned to design and simulate a fiber-optic link using OptiSystem.
- Understood link budget calculation and optical power losses.
- Observed the effect of attenuation on the received signal.
- Gained knowledge about BER analysis and signal quality.

RUBRICS:

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
System Design & DWDM Setup	Correct selection and connection of OptiSystem components for the DWDM link (PRBS Generator, NRZ Pulse Generator, CW Lasers, WDM Multiplexer, Optical Fiber, EDFA, WDM Demultiplexer, PIN Photodetectors, Low Pass Filters, BER Analyzers).	All four transmitter channels, multiplexer, fiber, EDFA, demultiplexer and receiver chains are correctly selected, connected in sequence and configured without errors.	All components are present with minor connection or layout errors, but the DWDM link remains functional.	Component selection is partially complete; missing receiver chains or EDFA placement errors impede signal flow.	Incorrect component selection or improper setup leading to complete simulation failure.	20%
Parameter Configuration & Execution	Accurate setup of input parameters: Transmit Power (20 dBm), Center Frequency (193.1 THz), Channel Spacing (100 GHz), Fiber Length (50 km, 75 km, 100 km) and Loop Control (0).	All input parameters and fiber-length test cases are accurately set and executed, with and without the EDFA.	Test case parameters are mostly accurate with minor setting discrepancies.	Basic parameters are configured, but multiple fiber-length cases or the with/without EDFA comparison are missing.	Incorrect parameter settings or failure to run the required test cases.	20%
Measurement & Performance Monitoring	Accurate extraction and recording of Min BER, Max Q-Factor, received power, optical spectrum and eye diagrams for all channels and fiber lengths.	Precise data extraction across all fiber lengths, neatly tabulated with complete BER, Q-factor and power readings and clear spectrum/eye diagrams.	Data is collected with minor omissions in BER, Q-factor or power readings.	Partial data collection with incomplete tables or missing spectrum/eye diagram captures.	Inaccurate, incomplete or missing data and measurements.	20%
Amplification & Fiber Loss Analysis	Detailed evaluation of fiber attenuation, dispersion, EDFA gain and noise, and the effect of amplification on received power, Q-factor and BER.	Comprehensive analysis correctly relating fiber length, attenuation and EDFA compensation to received power, Q-factor and BER, with and without amplification.	Clear analysis identifying basic trends in power, BER and Q-factor relative to fiber length and EDFA use.	Basic attempt at result analysis without explaining physical mechanisms such as attenuation, dispersion or amplifier gain.	Erroneous or missing analysis of simulation results.	20%
Report & Conclusion	Identification of the benefit of EDFA amplification, achievable transmission distance, trade-offs (amplifier noise, channel spacing), structure and clarity of the report.	Report features clear spectra and eye diagrams, complete result tables and insightful conclusions identifying optimal operating parameters for the DWDM link.	Satisfactory report with relevant displays and a sound concluding summary.	Incomplete report with missing graphs, readings or weak conclusions.	Disorganized or incomplete report lacking required diagrams and valid conclusions.	20%

CONCLUSION :

Thus, the fiber-optic communication link was successfully designed and simulated using OptiSystem. The link budget and BER were analysed to evaluate the signal quality and system performance. The experiment helped in understanding optical power losses, attenuation and the reliability of fiber-optic communication systems.